

EGT Particle Scaling: Predicting $402\times$ Resonances and the 402 GeV Dark Matter WIMP at the LHC

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The Emergent Gravity Theory (EGT) asserts that all fundamental masses harmonically lock to the universal geometric amplification factor, $\mathbf{A}_{\text{EGT}} \approx 402.3$. This paper translates this principle into a definitive high-energy prediction for the Large Hadron Collider (LHC). Specifically, the theory requires the existence of a dark matter weakly interacting massive particle (WIMP) with a mass precisely scaled by the EGT factor: $\mathbf{m}_\chi = 402 \text{ GeV}$. Furthermore, EGT predicts enhanced particle production cross-sections (σ_{EGT}) at energies that are $402\times$ multiples of fundamental scales, governed by the scale-dependent coupling operator $\mathbf{C}$. These $402\times$ resonances represent a critical, unambiguous, and currently hidden signature, providing the highest-energy test for geometric connectivity and demanding a focused search in the **400 GeV** to **1 TeV** regime of LHC data.

I. INTRODUCTION: HARMONIC LOCKING AND NEW PHYSICS

The Standard Model (SM) successfully describes known particles and forces but fails to account for gravity, Dark Matter (DM), or Dark Energy (DE). The EGT provides a complete quantum-cosmic unification by proposing that gravity emerges from geometric connectivity, which introduces a fundamental amplification factor $\mathbf{A}_{\text{EGT}} \approx 402.3$.

A core principle of EGT is that this factor acts as a universal scale, causing all fundamental parameters to ***harmonically lock*** to the **402** scaling. This geometric scaling must manifest directly in high-energy collisions, modifying particle interaction cross-sections.

II. EGT DARK MATTER MASS PREDICTION

The most direct and non-tunable particle-level prediction from the EGT framework is the mass of the Dark Matter candidate particle (χ), which is assumed to be a WIMP.

The EGT requires the WIMP mass to be directly related to the fundamental geometric factor $\mathbf{A}_{\text{EGT}}$:

$$\mathbf{m}_\chi = \mathbf{A}_{\text{EGT}} \cdot (1 \text{ GeV fundamental scale}) \quad (1)$$

Using $\mathbf{A}_{\text{EGT}} \approx 402.3$, the prediction is fixed at:

$$\mathbf{m}_\chi \approx 402 \text{ GeV}$$

This prediction is fully consistent with the observed dark matter density $\rho_{\text{DM}} \approx 0.3 \text{ GeV/cm}^3$, which is mathematically derived in EGT using this 402 GeV WIMP mass.

III. LHC RESONANCE ENHANCEMENT

The EGT-Photon Coupling term in the complete Lagrangian is:

$$\mathcal{L}_{\text{EGT}} = \lambda(1 + 2R)e^{-R/3} F_{\mu\nu} \tilde{F}^{\mu\nu} \quad (2)$$

where $\mathbf{R}$ is a scale-dependent term, $\mathbf{F}_{\mu\nu}$ is the electromagnetic field tensor, and $\tilde{\mathbf{F}}^{\mu\nu}$ is its dual. This scale-dependent coupling modifies the Standard Model cross-sections (σ_{SM}) at high energies ($\mathbf{E}$), leading to enhanced cross-sections (σ_{EGT}).

The predicted enhancement in collision cross-sections is given by:

$$\sigma_{\text{EGT}} = \sigma_{\text{SM}} \times (1 + 2E/E_{\text{EGT}}) \quad (3)$$

The EGT energy scale ($\mathbf{E}_{\text{EGT}}$) is the energy at which the $402\times$ scaling effect begins to dominate. Consistent with the mass prediction, EGT predicts ***resonances at $402\times$ multiples of fundamental scales***, placing the most significant signature in the **400 GeV** region and its harmonics ($\approx 800 \text{ GeV}$, $\approx 1200 \text{ GeV}$, etc.).

A. Test Condition: The Hidden 402 GeV WIMP

The most accessible test is the direct production and detection of the **402 GeV** WIMP (χ) at the LHC. If this particle is produced with an enhanced cross-section and subsequently decays into missing transverse energy, it would create an observable excess over SM predictions in the **400 GeV** energy bin.

The key challenge is the current search sensitivity: the EGT signal is currently ***hidden*** within the Standard Model backgrounds. A targeted search for di-jet or photon + missing energy final states with a narrow resonance at **402 GeV** is the definitive experimental condition for EGT detection.

IV. CONCLUSION

The EGT provides a concrete, quantitative, and testable high-energy prediction. The geometric constant $\mathbf{A}_{\text{EGT}}$ fixes both a Dark Matter WIMP mass ($\mathbf{m}_\chi = 402 \text{ GeV}$) and dictates an enhancement of cross-sections at $402\times$ energy multiples. A dedicated high-precision search by the ATLAS and CMS collaborations

for new physics resonances in the **400 GeV** mass range

is the critical condition for the direct observational proof of geometric particle scaling.

[1] B. Tice Sr. and Quintilius, Emergent Gravity Theory: Ultramagnetic Connectivity and Quantum-Cosmic Unification (Unpublished Manuscript).

[2] ATLAS and CMS Collaborations, Searches for New Physics in the 400 GeV to 1 TeV mass range (Used for current LHC WIMP constraints).

[3] B. Tice Sr. and Quintilius, EGT Cosmic Scaling: Predicting the $w_{\text{DE}} = -\mathbf{0.9975}$ Dark Energy Equation of State (Unpublished Manuscript).